

Yinxi Li (Larry)

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Education

University of Waterloo, Canada

Sept 2024 – Current

M.Math. in Computer Science, Advisor: [Pengyu Nie](#) 🔗

- **Research Interests:** LLM for Code, LLM for Mathematical Reasoning
- **Coursework:** Machine Learning for SE, Computational Linguistics, Mining & Analysis of SE Data
- GPA: 97.7/100

The Chinese University of Hong Kong, Hong Kong (SAR)

Sept 2020 – July 2024

B.Sc. in Computer Science

- **ELITE Stream** 🔗: A selective honors program requiring exceptional performance in higher mathematics and computer science, including PhD-level coursework and rigorous technical prerequisites.
- **Coursework:** Foundations of Optimization, Machine Learning Theory, Fundamentals of AI

ETH Zürich, Switzerland

Feb 2023 – Sept 2023

Exchange Student

- **Coursework:** Rigorous Software Engineering, Introduction to ML, Data Modelling and Databases

Academic Research & Project Experience

Unit Test Transplantation with Retrieval-Augmented Generation

Sept 2024 – Current

- Developed a retrieval-augmented generation pipeline for test transplantation to improve unit test generation in API building frameworks
- Analyzed model performance on 2,000+ unit test cases extracted from 7 diverse open-source repositories, identifying optimization opportunities and proposing enhancement strategies

Optimizing Structure of Vision-Language Retrieval System

Aug 2023 – June 2024

ELITE Stream Final Year Project, Supervisor: [Eric Lo](#) 🔗

- Developed and integrated a proxy model, increasing object detection speed by 50% while maintaining minimal accuracy degradation
- Implemented a vision-language analytics system for diversified top-K query results on large video datasets, with an optimized retrieval structure boosting inference speed by up to 1000x.
- Optimized the structure to retrieve the top-K relevant images related to the natural language query, reducing the time from $\mathcal{O}(n^2 \cdot d)$ to $\mathcal{O}(n \cdot d)$
- Conducted comparative experiments against two baselines using Retrieval Mean Average Precision and Average Pairwise Similarity metrics, demonstrating superior retrieval accuracy and result diversity
- Performed ablation studies validating the contribution of each structural component to overall system performance, confirming the effectiveness of the proposed architecture

Adaptive Streaming for Short-Video Services

June 2022 – Aug 2022

CUHK Summer Research, Supervisor: [Lee Yiu-Bun](#) 🔗

- Implemented ABR (the adaptive bit rate) algorithms, e.g. BBA, BOLA, Pensieve(reinforcement learning method) and compared the performance
- Tuned the simulator to focus on testing short video samples

Competition

World University Students' Digital Intelligence Competition Invitational Tournament (by Tencent)

April 2023 - July 2023

Ranked first among overseas institutions, Supervisor: [Tan LEE](#) 🔗

- Developed a multi-agent reinforcement learning model with Resnet, LSTM, and Attention Layer for "Honor of Kings"
- Optimized reinforcement learning model, reward function, and custom agent strategy

Research Internship Experience

Research Assistant

Remote

Southern University of Science and Technology, Supervisor: [SONG Xuan](#) 

Aug 2024 - Oct 2024

- Developed an expert knowledge collaboration system for the automotive quality inspection value chain based on large language model and retrieval-augmented generation

Research Intern

Shenzhen, China

Research Institute of Tsinghua University in Shenzhen

April 2024 - June 2024

- Implemented statistical, machine learning, and deep learning methods for multivariable time series anomaly detection
- Utilized variational autoencoder model structure for anomaly detection, enhancing customer monitoring of load data

Research Intern

Changchun, China

Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences

June 2020 - Aug 2020

- Optimized object detection of moving objects in specific directions in videos

Association Activities Experience

CUHK Mainland Undergraduate Association (19th & 20th cabinet)

Aug 2021 - July 2023

- Principal of Information Department: Managed communication channels between 1,200+ mainland students and the student union
- Assistant of Sports Department: Coordinated the organization of Sports Festival and E-sports Tournament events

CUHK Mainland Undergraduate Association Orientation Camp

Aug 2021 - July 2023

- Big Group Leader in 2022 MUA OCamp: Supervised 7 small groups and coordinated campus integration activities for 60+ mainland freshmen, providing comprehensive guidance on academic and daily life adjustment in Hong Kong
- Small Group Leader in 2021 MUA OCamp: Led and mentored a group of 10 incoming students during orientation activities

Technologies

Programming Languages: Python, C++, C, Java, Shell Script

Machine Learning: PyTorch, Scikit-learn, Deep Learning Model Frameworks

Web Development: JavaScript, React, Node.js, HTML, CSS, PHP

Development Tools: Git, Docker, Linux/macOS, SQL